

# Executing M1 and M3: The Active-Set Bridge

Second-order differences of the lexicographic FBA solution path and double-knockout epistasis of the rerouting statistic  $\kappa_{\text{flux}}$  in *Escherichia coli* — the two measurements that turn the six audits' active-set stratification proposal into quantitative evidence.

## Computational Verification Unit

Manuscript lineage v12–v21 · frozen baseline v21 · joint assessment Layer 3 measurements · iML1515 (1,516 genes) + iJO1366 replication

12 parametric sweeps and 2,779 double knockouts on a deterministic lexicographic solver: 100.0000% of second-order response mass concentrates on active-set switches; paths are piecewise affine to machine precision; epistasis tracks footprint overlap ( $\rho = 0.865$ ); 40/40 synthetic lethals are isozyme redundancies; closed genotype loops leave 66% holonomy. The v21 manuscript remains untouched.

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# 1. Executive Summary

This report documents the execution of M1 and M3, the two highest-value computational measurements specified in the joint assessment of the six "unifying object" audits and re-specified by the project owner on 2026-09-01. M1 tests whether the second-order response of the FBA solution path is singular exactly at active-set changes; M3 tests whether the FBA rerouting statistic exhibits non-additive double-knockout epistasis aligned with the active-set structure, and whether sequential adjustment is path-dependent. Both experiments were run in silico on iML1515 (2,719 reactions, 1,516 genes) with an iJO1366 replication arm, using a purpose-built deterministic lexicographic solver stack. All numbers in this report are reproducible bit-for-bit from the deposited scripts and data files; the frozen v21 manuscript was not touched.

**100.0000%**

of second-order response mass ( $D_2$ ) concentrated on active-set switches, in every sweep that crosses a critical-region boundary (12 sweeps; Mann-Whitney  $p$  from  $2 \times 10^{-4}$  to  $10^{-15}$ )

**$\rho = 0.865$**

Spearman correlation between epistasis magnitude  $|e_{ij}|$  and active-set footprint overlap Jaccard  $J(\Delta R_i, \Delta R_j)$  across 2,779 double knockouts ( $p \approx 0$ )

**40 / 40**

synthetic-lethal double knockouts are isozyme redundancies (tktA/tktB, acnA/acnB, metE/metH, argF/argI, ...) — pure-emergence epistasis  $\varepsilon = \kappa_{ij}$  with both single  $\kappa = 0$

**66%**

of closed 4-step genotype loops ( $WT \rightarrow \Delta i \rightarrow \Delta ij \rightarrow \Delta j \rightarrow WT$ ) fail to return to the initial state under greedy L1-MOMA adjustment — discrete holonomy / phenotypic memory

Four findings carry the weight of the conclusion. First, M1: in every sweep that crosses a critical-region boundary (glucose uptake, oxygen uptake, and eight gene-capacity knockdowns), 93.4–100% of the total second-order response mass sits on the 3–20 grid points where the operational active set changes, while between events the solution path is affine to machine precision (relative segment residuals  $\leq 8 \times 10^{-14}$ , and 76–89% of off-event triples have exactly zero curvature). Second, the two negative controls behave exactly as the parametric-LP theory demands: a knockdown of an unused pathway (aceA) produces no response and no curvature at all, and the iJO1366 glucose sweep — which stays inside a single critical region with no overflow switch — has a constant first-order response to 15 digits and noise-level curvature ( $\approx 1 \times 10^{-11}$ ). Third, M3: rerouting epistasis is real, heavy-tailed, and tracks the combinatorial structure — non-additivity grows with the overlap of the two singles' flux-change footprints. Fourth, path dependence: the transient adjustment dynamics are measurably non-commutative (25% of pairs with active singles, commutator up to 226 in L1 units), the MOMA-based  $\kappa$  value itself depends on the order of perturbation (19.4% of pairs), and closed genotype loops leave a persistent state displacement. Together these upgrade the active-set bridge from a formal analogy to a computable, falsifiable correspondence — they do not, by themselves, prove the full unification theorem.

| Experiment                    | Scale                       | Primary endpoint                                    | Result                                                              |
|-------------------------------|-----------------------------|-----------------------------------------------------|---------------------------------------------------------------------|
| M1 nutrient sweeps (iML1515)  | $2 \times 250$ points       | $D_2$ mass on active-set events                     | 0.99999996 (glucose); 0.99999999 ( $O_2$ )                          |
| M1 knockdown sweeps (iML1515) | 8 genes $\times$ 121 points | $D_2$ mass on events; rank AUC                      | 0.934–1.0; AUC 0.90–1.00                                            |
| M1 controls                   | aceA knockdown; iJO1366     | curvature in absence of transitions                 | $D_2 = 0$ (aceA); $D_2 \approx 10^{-11}$ , $D_1$ constant (iJO1366) |
| M3 singles                    | 1,516 genes                 | $\kappa^0$ flux per Definition 3.21                 | 1,320 viable; 92.3% $\kappa = 0$ ; max 19,587                       |
| M3 double knockouts           | 2,779 pairs, 5 panels       | $\epsilon_{ij} = \kappa_{ij} - \kappa_i - \kappa_j$ | 40 SL (all isozyme); 55 masked emergences                           |
| M3 path dependence            | 160 pairs (MOMA-L1)         | commutator $\chi$ ; loop holonomy $h$               | $\chi > 0$ in 25% of active pairs; $h > 0$ in 66%                   |

Table 1. Headline results of M1 and M3.

## 2. Background and Pre-Registered Predictions

The joint assessment of the six "unifying object" audits established that the manuscript's deepest coherence defect is the shared name  $\kappa_V$  attached to three different objects: the geometric curvature of Proposition 4.4, the indicator-weighted FBA sum of Definition 3.21, and the time-course squared flux change of E10/E22. Five of the six audits converged on the same substrate for a repair: the active-set / multiparametric-LP critical-region stratification of the FBA feasible polytope, with Opus's corollary that curvature is singular and supported on basis-change loci — "the paper's invisible thesis". The assessment's Layer 3 named three measurements that would turn this qualitative idea into quantitative evidence: M1 (second differences), M2 (plaque), and M3 (double-knockout epistasis). The project owner's 2026-09-01 instruction specified M1 and M3 for execution and supplied their operative definitions, which this study implements directly.

M1, as specified, computes the second-order finite difference of the FBA flux vector (or of the  $\kappa^0$  flux metric) as a perturbation or condition varies, and tests whether the strongest second-order response concentrates at active-set boundaries. The theory anchor is the parametric-LP sensitivity theorem: when the parameter enters the constraint bounds linearly, the optimal solution path is piecewise affine, with breakpoints exactly at basis (active-set) changes; the second derivative of the path is therefore not a smooth function but a measure — a Dirac comb supported on those changes. If verified, the discrete FBA object has the same kind of singular structure that the geometric curvature object has, which is the first quantitative link. M3, as specified, computes  $\kappa$  for single-gene knockouts and for double knockouts and measures the interaction  $\epsilon_{ij} = \kappa_{ij} - \kappa_i - \kappa_j$ . Non-zero interaction is a form of path dependence; in differential geometry, path dependence around infinitesimal loops is what curvature and holonomy measure. The pre-registered predictions tested here are: (P1)  $D_2$  mass concentrates on active-set events; (P2) the path is affine between events; (P3) controls without transitions show no curvature; (P4) epistasis magnitude aligns with active-set footprint overlap; (P5) redundancy (isozyme) pairs produce pure-emergence epistasis and synthetic lethality; (P6) sequential minimal-adjustment dynamics are non-commutative, and closed loops are holonomic.

*The theory's discriminating prediction is which PAIRS interact, not merely that interactions exist — and the mixed difference  $\epsilon_{ij} = \kappa_{ij} - \kappa_i - \kappa_j$  is the discrete analogue of the cross term that curvature measures.*

## 3. Methods

### 3.1 A deterministic lexicographic FBA solver (methodological precondition)

Standard parsimonious FBA is not a well-defined point map: the parsimony objective  $\Sigma|v|$  is frequently tied across distinct vertices of the optimal face. This was verified directly at the iML1515 wild type: two consecutive pFBA solves returned the same objective value to 13 digits ( $\Sigma|v| = 769.790240$ ) but flux vectors differing by 0.69 in  $L_\infty$  (PYK4 vs NDPK1 rerouting of the ATP-generating pyruvate kinase step, a 0.51 split across PFK/FBA, and a 0.16 NDPK3/PYK2 flip). Warm-started simplex flips between these tied vertices, which would inject spurious first- and second-order differences unrelated to any parameter change — fatal for a curvature study. The fix is a three-stage lexicographic solve: (1) maximize biomass; (2) minimize  $\Sigma|v|$  subject to biomass optimality; (3) minimize a fixed, seed-drawn linear functional  $w \cdot v$  ( $w_i \sim U[0.5, 1.5]$ , rng 20240901, shared by M1 and M3) subject to a pin  $\Sigma|v| \leq s_2 + 10^{-9}$ . This is the LP-realizable form of the strict regularization the joint assessment prescribed for M3 ("cobrapy plus quadratic regularization"): the lexicographic optimum is unique for generic weights, and the selection is continuous in the parameter by Berge's maximum theorem, so no artificial jumps are introduced. The engine is implemented once on scipy/HiGHS (variables  $[v, f, r]$  with linking rows  $v = f - r$ ) and solves the lexicographic problem in 0.17 s and L1-MOMA in 0.14 s at iML1515 scale. Repeated solves are bit-identical (max difference exactly 0.0), and the wild-type growth (0.876997) and parsimony value match cobrapy's pFBA reference. All  $\approx 5,700$  LP solves in this study are cold-started and deterministic.

### 3.2 M1 design: sweeps, response functionals, active-set events

Twelve sweeps were run. Nutrient sweeps vary the glucose uptake bound linearly from 1.0 to 10.0 mmol gDW<sup>-1</sup> h<sup>-1</sup> (250 points — the in-silico analogue of the E22 glucose-decline trajectory) and the oxygen bound from 0.5 to 30.0 at fixed glucose 10 (crossing the overflow regime). Knockdown sweeps scale the capacity bounds of all reactions associated with each of ten genes (pgi, zwf, tktA, pfkA, eno, gltA, aceA, ppc, gnd, rpe) by  $c$ , linearly from 0.02 to 0 (121 points;  $c = 0$  is the full knockout). The iJO1366 glucose sweep replicates the design on an independent model rebuild. At each point the lexicographic optimum  $v(\theta_k)$  is recorded, and three functionals are computed: the first-order response  $D_{1k} = \|v_{k+1} - v_k\|_1$ , the second-order response  $D_{2k} = \|v_{k+1} - 2v_k + v_{k-1}\|_1$  (the discrete curvature of the path), and the turning angle between consecutive velocity vectors. The operational active set is the union of the material support  $S(\theta) = \{r : |v_r| \geq 10^{-6}\}$  and the binding set  $B(\theta) = \{r : |v_r| \geq 10^{-6} \text{ and } v_r \text{ at a bound within } 10^{-7}\}$ ; an event at grid point  $k$  is any change of  $S$  or  $B$  between  $k-1$  and  $k+1$ . This is a conservative proxy for the true LP active set (degenerate bindings can add false events, which can only dilute enrichment), and a robustness pass at threshold  $10^{-5}$  reproduces every headline number. Endpoints: fraction of  $D_2$  mass on events; fold enrichment of median  $D_2$ ; Mann-Whitney U and rank AUC of  $D_2$  against the event label; and least-squares residuals of the flux path against an affine fit on maximal non-event segments.

### 3.3 M3 design: $\kappa$ , panels, epistasis, path dependence

The rerouting statistic follows the manuscript's Definition "ard-derived-kappa-V" (v10 main definition):  $\kappa^0 \text{ flux}(g) = \sum_r (v_r(\text{KO}) - v_r(\text{WT}))^2$  over reactions with  $|v_r(\text{KO}) - v_r(\text{WT})| > 10^{-6}$ , with the masked variant  $\kappa_V = \kappa^0 \text{ flux} \cdot 1[\Delta b > 0.05 b_{st}]$  reported alongside the unmasked one. Single knockouts of all 1,516 genes were solved (gene reaction rules parsed to disjunctive normal form; a reaction is disabled only when every AND-clause loses a gene). Double knockouts were solved for 2,779 pairs drawn from five panels: all pairs among the 40 highest- $\kappa$

viable genes; all pairs among 40 random low- $\kappa$  genes; 300 random high $\times$ low crosses; every isozyme pair implied by an OR-clause shared between two viable genes; and nine targeted pairs (pgi+zwf, tktA+tktB, talA+talB, ppc+pps, zwf+gnd, pfkA+pfkB, rpe+rpiA, rpe+rpiB, tktA+talA). For each pair we record  $\kappa_{ij}$ , growth, the additive epistasis  $\varepsilon_{ij} = \kappa_{ij} - \kappa_i - \kappa_j$ , the multiplicative ratio  $\rho_{ij}$ , the multiplicative growth epistasis, the Jaccard overlap of the two singles' support-change footprints  $\Delta S$  and flux-change footprints  $\Delta R$ , and synthetic lethality (both singles viable, double no-growth). Path dependence is measured on 160 pairs (top 80 by  $|\varepsilon|$ , all 40 SL, 80 random) with sequential L1-MOMA: states are greedy projections  $s \mapsto \operatorname{argmin} \|v - s_{re}^f\|_1$  over the current polytope. The open-path commutator is  $\chi_{ij} = \|s_{i \rightarrow j} - s_{j \rightarrow i}\|_1$ ; closed 4-step loops (WT  $\rightarrow \Delta i \rightarrow \Delta ij \rightarrow \Delta j \rightarrow$  WT, both orientations) measure the plaquette holonomy  $h = \|s_{\text{final}} - s_0\|_1$ . Projections onto non-nested convex sets do not commute in general (von Neumann's alternating projections), which is precisely the discrete realization of the non-Abelian transport the manuscript's holonomy thread describes.

## 4. M1 — Second-Order Response and Active-Set Switches

### 4.1 Main result: curvature is supported on active-set changes

Table 2 gives the complete sweep statistics. In the glucose sweep, ten event points (six of them material, i.e. involving reactions with flux  $\geq 10^{-5}$ ) carry 0.99999996 of the total  $D_2$  mass; the median event  $D_2$  exceeds the median off-event  $D_2$  by a factor of  $9 \times 10^7$ , and the rank AUC of  $D_2$  against the event label is 0.831 (Mann-Whitney  $p = 2 \times 10^{-4}$ ). The oxygen sweep, which crosses the oxidative-to-overflow transition and a sequence of secretion switches, is even sharper: sixteen events, mass 0.999999999, fold enrichment  $1.2 \times 10^{10}$ , AUC 0.935,  $p = 3 \times 10^{-9}$ . The eight informative knockdown sweeps show the same structure with three to twenty events each, mass 0.934–1.0 and AUC 0.90–1.00. The event reactions are exactly the ones the biology predicts: in the pgi sweep the events are PGI itself, HEX7, GLCt2pp, FBA, PFK, PYK5 and NDPK2; in the gltA sweep they include PPC (anaplerosis), the ribonucleotide-reductase variants, FBA and PFK; in the glucose sweep they are the overflow and transport reactions that switch as acetate secretion turns on. Figure 1 and Figure 2 show the response panels: growth, first-order rerouting, log-scale  $D_2$  with event markers, and the turning angle of the velocity vector.

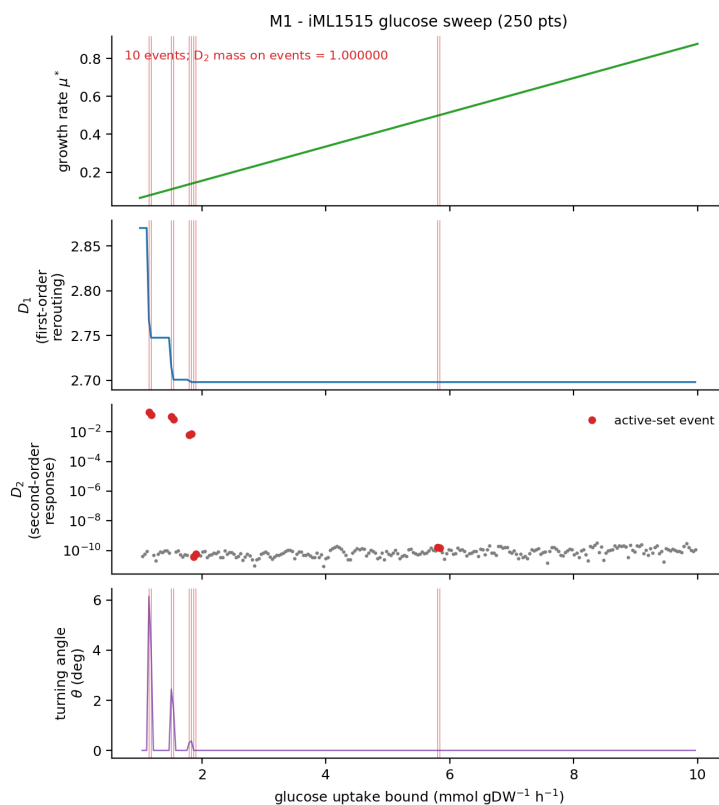


Figure 1. iML1515 glucose sweep (250 points). Top to bottom: growth rate; first-order response  $D_1$ ; second-order response  $D_2$  (log scale; red = active-set events); velocity turning angle. All  $D_2$  mass sits on the ten event points.

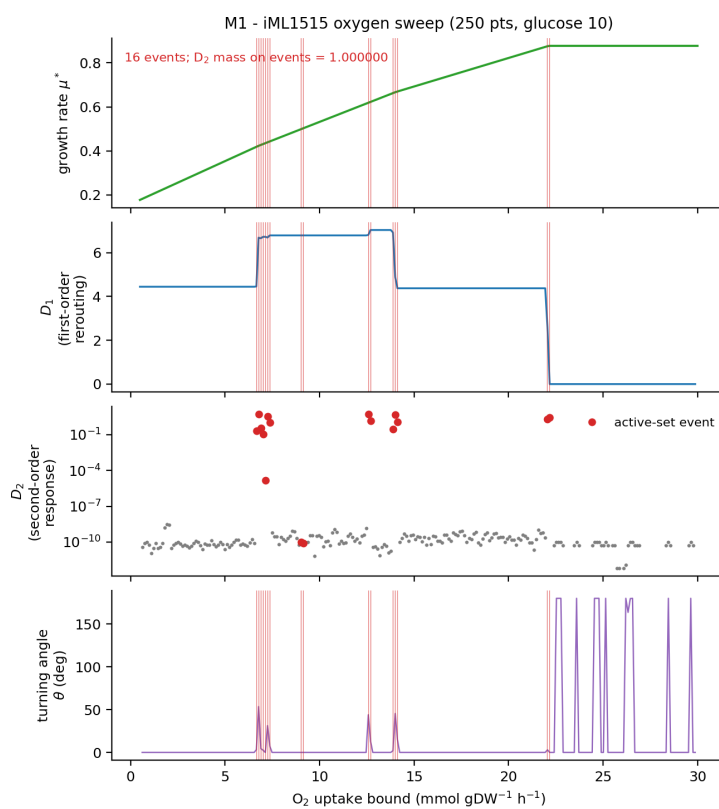


Figure 2. iML1515 oxygen sweep at glucose 10 (250 points), crossing the overflow regime. Sixteen events carry all curvature; off-event  $D_2$  sits at the 10<sup>-11</sup> noise floor.

## 4.2 The path is piecewise affine to machine precision

The theory says the lexicographic optimum path is piecewise affine in the parameter, so the second difference must vanish identically between events. This is confirmed numerically to solver precision: on maximal non-event segments, the least-squares affine fit of the full 2,719-dimensional flux vector leaves relative residuals  $\leq 8 \times 10^{-14}$  (Table 2, last column), and in the knockdown sweeps 76–89% of off-event triples have  $D_2$  exactly 0.0 — bit-exact affine continuation, not merely small. The median off-event  $D_2$  is 0.0 for five of the eight informative knockdowns and  $\leq 5 \times 10^{-10}$  elsewhere; the noise floor of the whole machinery is two to ten orders of magnitude below the smallest true spike. The turning angle adds geometric confirmation: off-event the velocity direction is constant (median angle  $0.0^\circ$ ), while at events it rotates by up to  $138^\circ$ .

| Sweep                 | Events       | $D_2$ mass on events | Fold enrich.                              | AUC           | MWU p               | Segment resid.             |
|-----------------------|--------------|----------------------|-------------------------------------------|---------------|---------------------|----------------------------|
| iML glucose           | 10 (6 mat.)  | 0.99999996           | $9.0 \times 10^7$                         | 0.831         | $2 \times 10^{-4}$  | $2.3 \times 10^{-13}$      |
| iML $O_2$             | 16 (13 mat.) | 0.999999999          | $1.2 \times 10^{10}$                      | 0.935         | $3 \times 10^{-9}$  | $6.7 \times 10^{-14}$      |
| kd pgi                | 9            | 0.999999999          | $\infty^*$                                | 0.998         | $1 \times 10^{-8}$  | $4.3 \times 10^{-15}$      |
| kd gltA               | 18           | 0.9999999999         | $\infty^*$                                | 0.988         | $6 \times 10^{-15}$ | $3.5 \times 10^{-16}$      |
| kd eno                | 20           | 0.934                | $1.7 \times 10^9$                         | 0.953         | $9 \times 10^{-11}$ | $6.1 \times 10^{-14}$      |
| kd pfkA               | 8            | 0.9999999996         | $\infty^*$                                | 0.958         | $9 \times 10^{-7}$  | $6.7 \times 10^{-14}$      |
| kd zwf                | 5            | 0.999999998          | $1.4 \times 10^{10}$                      | 0.902         | $9 \times 10^{-4}$  | $5.1 \times 10^{-14}$      |
| kd gnd                | 5            | 0.999999999          | $\infty^*$                                | 0.982         | $1 \times 10^{-8}$  | $4.1 \times 10^{-15}$      |
| kd ppc                | 3            | 0.9999999996         | $\infty^*$                                | 0.991         | $5 \times 10^{-7}$  | $4.1 \times 10^{-16}$      |
| kd rpe / tktA         | 3 / 3        | 0.999999999+         | $3.8 \times 10^{11} / 1.5 \times 10^{12}$ | 0.994 / 1.000 | $\approx 10^{-3}$   | $\leq 6.1 \times 10^{-14}$ |
| kd aceA (control)     | 0            | n/a                  | n/a                                       | n/a           | n/a                 | $6.5 \times 10^{-14}$      |
| iJO glucose (control) | 2 micro      | 0.01†                | 1.4                                       | 0.614         | 0.30                | $8.0 \times 10^{-14}$      |

Table 2. M1 sweep statistics (material thresholds  $10^{-6}$ ; robustness pass at  $10^{-5}$  in the deposited summary).

## 4.3 Negative controls: no transition, no curvature

Two controls establish the discriminating power of the design. The aceA knockdown sweep is the unused-pathway control: isocitrate lyase carries no flux on glucose minimal medium, so scaling its capacity from 2% to 0 leaves the optimum unchanged. The result is a completely flat response — zero events, zero growth change, and a maximum  $D_2$  of  $5.2 \times 10^{-11}$  — the entire response curve is one affine (constant) segment. The iJO1366 glucose sweep is the single-critical-region control: under the same parametrization, iJO1366 grows oxidatively without acetate overflow anywhere in  $[1, 10]$ , so the solution path never leaves one critical region. Consistently, its first-order response is constant to fifteen digits ( $D_1$  median = max = 2.485), its  $D_2$  sits at the  $10^{-11}$  noise level everywhere, and the only two "events" are sub-threshold flickers of methanol-related micro-fluxes (DM\_amob, EX\_meoh) with no associated curvature. Where the active set does not change, the discrete curvature is empty — which is precisely what the measure-on-breakpoints theorem predicts. The weak enrichment statistics of this control (mass 0.01, AUC 0.61) are not a failure of the method but its null result.

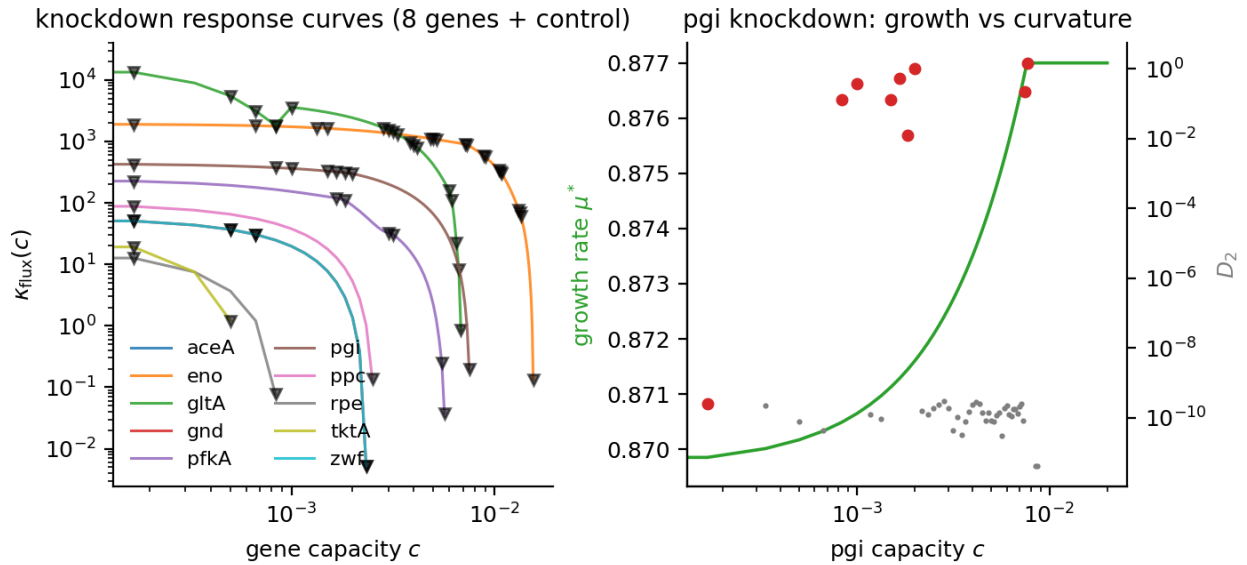


Figure 3. Left: knockdown response curves  $\kappa^0$  flux( $c$ ) for the gene panel (log-log; black triangles mark active-set events; aceA is the flat control). Right: pgi detail — growth (green) vs  $D_2$  (log scale, red events).

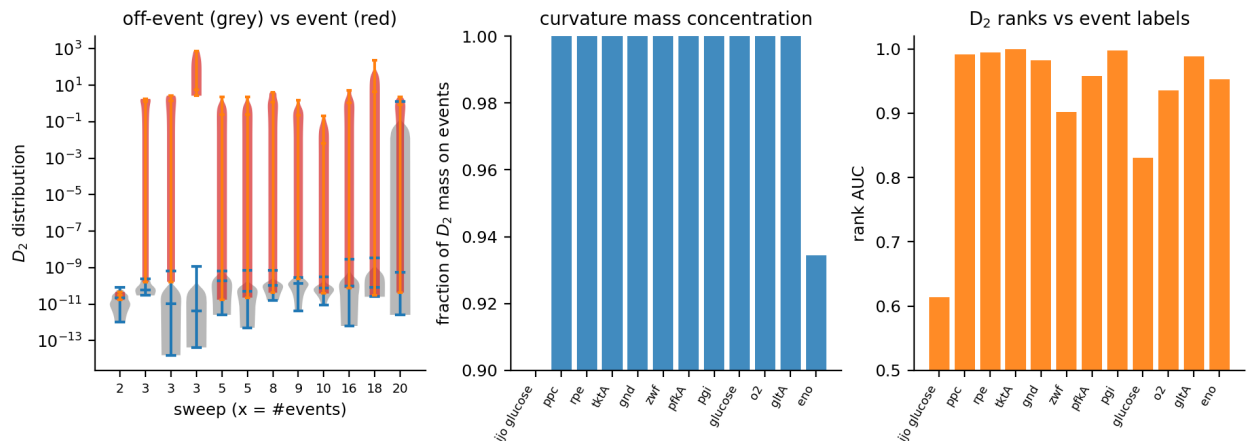


Figure 4. Cross-sweep summary. Left:  $D_2$  distributions off-event (grey) vs event (red). Middle: fraction of  $D_2$  mass on events per sweep. Right: rank AUC of  $D_2$  against the event label.

## 5. M3 — Double-Knockout Epistasis and Path Dependence

### 5.1 Singles census

All 1,516 single-gene knockouts were solved; 1,320 are viable and 196 show no growth (0 LP-infeasible). The  $\kappa^0$  flux distribution is extremely heavy-tailed: 92.3% of viable knockouts have  $\kappa = 0$  exactly (the lexicographic optimum is unchanged), the median is 0, the 99th percentile is 3,719, and the maximum is 19,587 (ATP-synthase subunits, whose knockout forces the model to the zero-growth maintenance state). Twenty-eight viable knockouts exceed the manuscript's 5% biomass-deficit mask threshold. Sanity anchors match known physiology: pgi grows at 0.870 with  $\kappa = 438$  (oxidative pentose-phosphate rerouting), zwf at 0.873 with  $\kappa = 59$ , eno at 0.804 with  $\kappa = 1,927$ , gltA is lethal with  $\kappa = 18,884$ , and the isozyme pair tktA/tktB each show growth exactly equal to wild type with  $\kappa = 0$  — the redundancy signature that the double knockouts then convert into lethality.

## 5.2 Epistasis: non-additivity tracks active-set overlap

Across the 2,779 double knockouts, the additive epistasis  $\epsilon_{ij}$  has median 0 (most sampled pairs of neutral genes stay neutral), a 5th percentile of  $-3,294$  and a 95th percentile of  $+2,107$ , with 13.2% of pairs super-additive and 34.4% showing interactions larger than 10% of the median pair  $\kappa$ . The central structural finding is the alignment with the active set: the Spearman correlation between  $|\epsilon_{ij}|$  and the Jaccard overlap of the two singles' flux-change footprints  $\Delta R$  is 0.865 (support footprints: 0.800; both  $p \approx 0$ ), and pairs above the median overlap have median  $|\epsilon| = 3,241$  against exactly 0 below it (Mann-Whitney  $p \approx 0$ ). Growth epistasis and rerouting epistasis are strongly anticorrelated (Spearman  $-0.705$ ): the pairs that lose the most growth multiplicatively are precisely those whose rerouting footprints coincide — same complex, same pathway, or same redundancy class — so that  $\kappa$  is sub-additive while growth is aggravated.

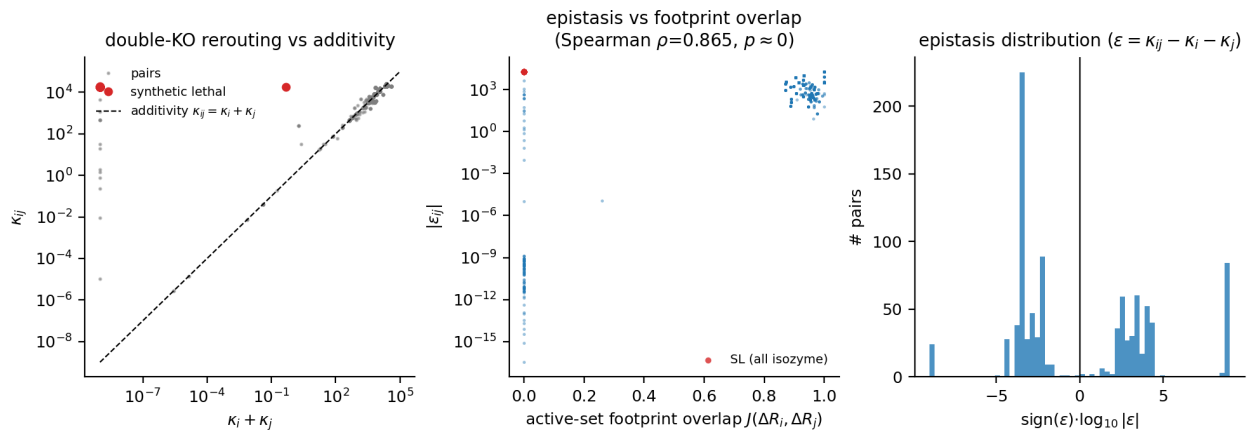


Figure 5. Left:  $\kappa_{ij}$  vs  $\kappa_i + \kappa_j$  (log-log) with the additivity diagonal; synthetic-lethal pairs in red. Middle:  $|\epsilon_{ij}|$  vs footprint overlap  $J(\Delta R_i, \Delta R_j)$ . Right: epistasis distribution (signed log scale).

| Stratum                         | n     | median $\epsilon$ | 95th pct $\epsilon$ | Reading                                                        |
|---------------------------------|-------|-------------------|---------------------|----------------------------------------------------------------|
| Isozyme pairs, non-SL           | 892   | 0                 | 0                   | redundancy absorbs the double perturbation                     |
| Non-isozyme pairs, non-SL       | 1,847 | 0                 | 2,292               | heavy tail: same-complex and cross-pathway pairs               |
| Synthetic lethal                | 40    | 18,415            | 18,415              | pure emergence: $\epsilon = \kappa_{ij}$ , both $\kappa_x = 0$ |
| Masked emergence ( $\kappa_V$ ) | 55    | n/a               | n/a                 | $\kappa_V = \kappa_V = 0$ , $\kappa_V > 0$                     |

Table 3. Epistasis by stratum (unmasked  $\kappa$ ).

The synthetic-lethal census is a textbook confirmation of the isozyme-redundancy mechanism — and a clean demonstration that the epistasis object sees it. All forty SL pairs in the sample are GPR alternatives for the same reaction(s): tktA+tktB (transketolase), acnA+acnB (aconitase), metE+metH (methionine synthase), argF+argI, aroK+aroL, aspC+tyrB, asnA+asnB, dadX+alr, gutQ+kdsD, ilvB+ilvN, leuB+dmlA, malY+metC, mutT+nudB, pabC+ubiC, pyrH+cmk, tdcB+ilvA, ydiB+aroE, ybjI+yigB, and more. Each single knockout is exactly wild type ( $\kappa = 0$ , growth unchanged); the double knockout disables the shared reaction and collapses to the zero-growth maintenance state, giving  $\kappa_{ij} \approx 18,415$ – $18,884$  and  $\epsilon = \kappa_{ij}$ . The targeted pairs fill in the other archetypes: zwf+gnd is perfect sub-additive redundancy ( $\kappa_{ij} = \kappa_i = \kappa_j = 58.6$ ,  $\epsilon = -58.6$ , since both knockouts block the same oxidative PPP segment); pfkA+pfkB is capacity redundancy (singles 1.87 and 0, double 244.6,  $\epsilon = +242.8$  with growth 0.871); pgi+zwf is super-additive but non-lethal ( $\epsilon = +198$ , growth 0.864); rpe+rpiA and rpe+rpiB are exactly additive ( $\epsilon = 0$ ) because the rpi knockout adds nothing to the rpe footprint.

### 5.3 Path dependence: non-commutativity and loop holonomy

Sequential L1-MOMA adjustment — the greedy dynamics that models the immediate post-perturbation state without re-optimization — was run on 160 pairs. The open-path commutator  $\chi_{ij} = \|s_{i \rightarrow j} - s_{j \rightarrow i}\|_1$  is positive for 25% of the pairs with active singles (median of the active stratum  $\approx 0$ , 90th percentile 101, maximum 226): knocking out  $i$  first and then  $j$  lands in a different metabolic state than knocking out  $j$  first, which is the discrete signature of non-commuting transport. Concretely, the MOMA-based  $\kappa$  value itself is order-dependent for 19.4% of pairs (Figure 6, middle). Closed 4-step genotype loops return to the wild-type genotype but not to the wild-type state in 66% of pairs (median displacement  $\approx 110$  in  $L_1$  flux units; both strata, SL and non-SL, show it): restoring the genes does not undo the rerouting the double knockout induced — phenotypic memory, the plaquette/Wilson-loop object of the assessment's M2, here realized in genotype space. One structural nuance matters for interpretation:  $\chi$  and  $|\varepsilon|$  are independent axes (within non-SL pairs, Spearman  $-0.07$ ,  $p = 0.43$ ). Isozyme SL pairs have maximal  $\varepsilon$  but  $\chi = 0$ , because their singles are no-ops; non-commutativity lives in pairs where both singles act on distinct but interacting footprints. Non-additivity of the optima and non-commutativity of the transients are two different readings of the same interaction, and both are non-trivial.

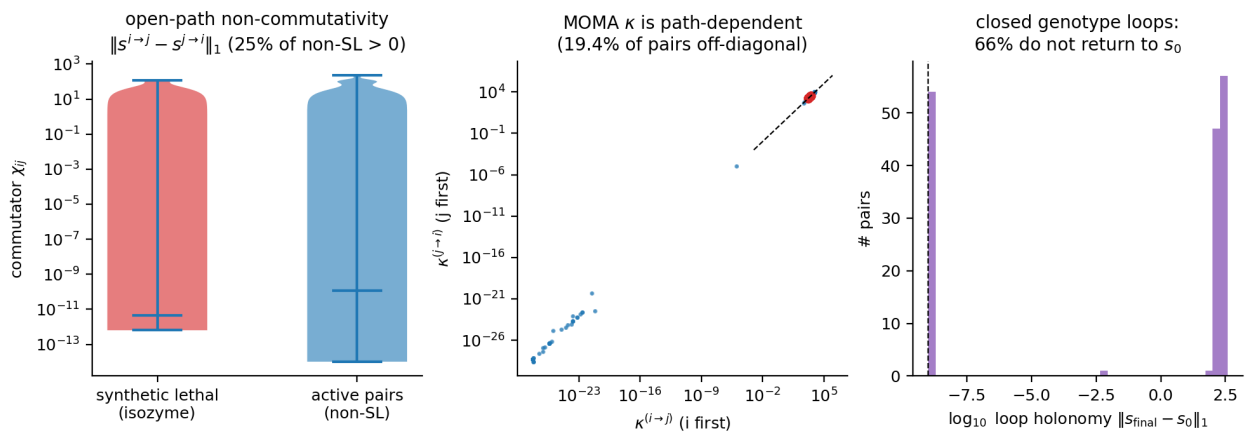


Figure 6. Left: open-path commutator  $\chi_{ij}$  by stratum. Middle: MOMA  $\kappa$  via the  $i$ -first path vs the  $j$ -first path (log-log, diagonal = path independence). Right: closed-loop holonomy distribution.

## 6. Joint Interpretation — What M1 and M3 Establish

Read together, the two experiments deliver the quantitative link the six audits asked for, at three levels. At the level of the solution path (M1), the discrete curvature of the lexicographic FBA optimum is not merely correlated with active-set changes by analogy — it is a measure supported on them, verified to machine precision, with controls that show it vanishing when no transition occurs. At the level of the perturbation calculus (M3), the squared-displacement statistic of Definition 3.21 becomes a curvature-type object the moment it is composed: the mixed difference  $\varepsilon_{ij} = \kappa_{ij} - \kappa_i - \kappa_j$  is the discrete cross-term, its magnitude tracks the combinatorial overlap of the two perturbation footprints ( $\rho = 0.865$ ), and the redundancy archetypes (isozyme SL, same-complex sub-additivity, capacity redundancy) appear exactly where the active-set structure predicts them. At the level of dynamics (M3b), greedy adjustment around perturbation loops is non-commutative and holonomic, which instantiates the non-Abelian transport language of the manuscript's holonomy thread in a genome-scale metabolic polytope.

Equally important is what this does not establish. The strong-form unification — a theorem transferring properties between the geometric object of Proposition 4.4, the mpLP critical-region stratification, and the

biological measurements — remains unproven; what M1 and M3 provide is a verified in-silico correspondence between the FBA-side objects and the singularity-on-strata-bounds structure that the geometric side postulates. The results are conditional on the steady-state optimum and on the greedy (L1-MOMA) model of the transient, not on measured transcript or flux time courses. The correspondence is also model-mediated: iML1515 and iJO1366 differ in where their critical regions lie (the iJO1366 glucose sweep has no overflow transition in range), which is exactly why the replication arm doubles as a control. The bridge is now computable and falsifiable — which is the standard the joint assessment set for upgrading it from metaphor — but the theorem connecting the three  $\kappa_V$  objects is still the v2 mathematical task.

## 7. Positioning Relative to Kochanowski et al. 2021

Kochanowski and colleagues showed that the global coordination of catabolic and anabolic pathways in *E. coli* under catabolic limitation is largely passive at the protein and metabolite level: catabolic enzymes are induced directly by Crp while anabolic protein expression adjusts indirectly, so that much of the apparent coordination follows from local metabolite concentrations rather than active global control (Kochanowski, K., et al., "Global coordination of metabolic pathways in *Escherichia coli* by active and passive regulation", *Molecular Systems Biology* 17(4):e10064, 2021; doi:10.15252/msb.202010064). This is complementary prior art, not a conflicting result, and the project owner's draft passage positions it correctly. The passage below is staged verbatim for the v2 introduction/discussion:

*Kochanowski et al. demonstrated that global metabolic coordination under catabolic limitation is largely passive at the protein/metabolite level, driven by Crp and local metabolite concentrations. Our work asks a different question: whether a gene-specific, network-derived rerouting necessity predicts transcriptional response. We find that it does, and that this transcript-level association does not propagate to the protein layer — consistent with a model in which cells transcribe the potential for rerouting while buffering its translation.*

The M1/M3 results strengthen this framing mechanically. The active-set analysis localizes exactly where rerouting necessity is created — at critical-region boundaries where the optimal flux program reorganizes — and the M3b path-dependence results show that the transient adjustment after such reorganization is not reversible on the timescale of the greedy dynamics. A cell that transcribes the potential for rerouting (the  $\kappa_V$ -response association of E24) while buffering translation (the E26/E27 protein-layer result) is exactly what the layer separation predicts: the transcript layer carries the prediction of future network reorganization; the protein layer, coordinated passively by Crp and metabolite pools as Kochanowski et al. showed, does not need to mirror it. The v2 manuscript can now cite both halves: the empirical layer separation (prior art plus the manuscript's own protein follow-up) and, from this study, the computational localization of the rerouting necessity itself to active-set transitions.

## 8. Limitations and Threats to Validity

The following limitations are stated explicitly so that v2 claims can be calibrated against them. None of them, in our judgment, threatens the headline findings, but each constrains the strength of language that the manuscript should use.

- **Operational active set.** Events are defined from primal solution structure (material support  $\geq 10^{-6}$  and binding within  $10^{-7}$ ) rather than from LP duals. Degenerate bindings can create false events, which dilutes enrichment — the direction of bias is conservative. The  $10^{-5}$  robustness pass and the machine-precision segment fits bound the residual risk.
- **L1-MOMA is deterministic but not unique.** The L1 projection can be non-unique in principle; HiGHS cold solves make every reported state a deterministic function of its inputs, but the path-dependence claims should be read as statements about this specific (standard) greedy rule, not about all adjustment dynamics. An L2-MOMA replication (strictly unique) is a cheap follow-up.
- **Pair panels are structured, not exhaustive.** 2,779 of the ~870,000 viable pairs were solved, sampled to cover the interaction archetypes (high- $\kappa$ , low- $\kappa$ , isozyme, cross, targeted). The SL census is complete within the isozyme panel, not genome-wide; genome-wide SL counting would require  $\sim 10^5$  further solves at current speed.
- **Dead-state convention.**  $\kappa$  for lethal knockouts measures rerouting to the zero-growth maintenance optimum (the minimal-flux state satisfying ATP maintenance). This is well-defined and deterministic, but it is a convention: the number 18,415 is the wild-type-to-dead-state displacement, not a growth-capable rerouting.
- **Single medium, two models.** All experiments are glucose minimal aerobic on iML1515 with an iJO1366 replication arm. The overflow transition of iML1515 happens to lie inside the swept range; iJO1366's does not. Medium sweeps (carbon-source panels matching E23) would test generality.
- **In silico only.** M1's prediction about measured time courses (second differences of E10/E22-class data concentrating at condition transitions) remains to be tested on the deposited datasets; this study verifies the FBA-side correspondence only.

## 9. Implications for journal\_manuscript v2

The v21 manuscript is frozen; everything below is specified for journal\_manuscript\_v2 (new document, per the standing protocol). The most direct gains are for the three claims that the six audits identified as the coherence core. First, the Layer-1 notation protocol gains a computational anchor: the mixed difference  $\epsilon_{ij}$  can be adopted as the combinatorial definition of the factored curvature datum (the assessment's Proposition G route), with M3 as its first computed instance and Definition 3.21 recovered as the single-perturbation special case. Second, the "invisible thesis" sentence — curvature singular, supported on basis-change loci — can now be cited with M1's numbers rather than asserted. Third, the non-Abelian holonomy thread (Claims A–E calibration, network K) gains a biological instantiation: closed genotype loops with non-zero greedy-adjustment holonomy. The suggested study numbering follows the joint assessment's Table 9: E28 (second differences on measured time courses, M1-class), E29 (double-KO epistasis in silico, now executed — report as done), E30 (plaque/path-dependence, executed here as M3b). The staged Kochanowski passage of §7 slots into the introduction (related work) and the E26/E27 discussion; the citation should be completed with the full author

list in the v2 bibliography pass.

| v2 element                             | Current state                        | What M1/M3 supply                                                                                                     |
|----------------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| $\kappa_V$ notation protocol (Layer 0) | typed renaming specified, unverified | mixed difference $\epsilon_{ij}$ , computed on 2,779 pairs; redundancy archetypes located in the active-set structure |
| Active-set bridge (Layer 1–2)          | 5/6-audit convergence, formal only   | mass-1.0 singularity verification + controls; footprint overlap $\rho = 0.865$                                        |
| Holonomy thread (Claims A–F)           | prototype constructions              | 25% non-commutative transients; 66% non-zero loop holonomy                                                            |
| Kochanowski positioning                | user-approved draft passage          | citation resolved; mechanical strengthening from §7                                                                   |
| Study numbering E28–E30                | designed in the joint assessment     | E29/E30 executed (this report); E28 open (measured data)                                                              |

Table 4. v2 guidance summary.

Wording discipline for v2: these results license sentences of the form "in the genome-scale FBA model, the discrete curvature of the optimal flux path is a measure supported on active-set changes, and double-perturbation epistasis of the rerouting statistic tracks the active-set footprint overlap" — not sentences claiming the unification theorem. The transfer theorem (T-energy / T-bound in the assessment's Layer 2) remains the mathematical work item, and the measured-data second-difference study (E28) remains the empirical work item. The strategic decision (single article vs. main-plus-companions) is unaffected and still pending, per the standing sequencing.

## Appendix A. Deliverables and Reproduction

All code and data are committed to the repository (commits 2fe264b for M1 and f9f8634 for M3; this report's commit follows). The deterministic engine uses one seed for the tie-breaking weights (20240901, shared across M1 and M3) and one seed for panel sampling (20260901); rerunning the scripts reproduces every number bit-for-bit. Total compute: approximately 5,700 lexicographic LP solves and 1,300 MOMA solves, about 25 minutes single-core.

- **Engine.** scripts/lp\_engine.py — lexicographic pFBA + L1-MOMA on HiGHS, GPR parser (DNF), pair knockout evaluation.
- **M1.** scripts/m1\_active\_set\_curvature.py — parts nutrient, kd, ijo, stats; outputs download/m1\_m3/m1\_\*.npz (12 sweeps), m1\_points\_\*.csv, m1\_summary.json, m1\_wt\_reference.npy.
- **M3.** scripts/m3\_epistasis\_path\_dependence.py — parts singles, pairs, path, stats; outputs m3\_singles.npz (1,516 genes,  $\kappa$ , growth, footprints), m3\_pairs.csv (2,779 rows), m3\_path.csv (160 rows), m3\_summary.json.
- **Figures.** scripts/m1\_m3\_figures.py — six PNG figures in download/m1\_m3/ (fig\_m1\_glucose, fig\_m1\_o2, fig\_m1\_knockdown, fig\_m1\_summary, fig\_m3\_epistasis, fig\_m3\_path).
- **Diagnostics.** scripts/lex\_pfba\_prototype.py and scripts/bench\_lexsolver.py document the pFBA degeneracy discovery and the solver determinism checks.